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337512 (37)

BE (5th Semester)

Examination, Nov.-Dec., 2014

Branch : Mechanical

INTERNAL COMBUSTION ENGINES

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all five questions. Part 'a' is compulsory. Attempt any two from 'b', 'c' and 'd'. Explain your answer with neat sketches where required. All question carry equal marks.

Q. 1. (a) Draw p-v diagram for SI and C-I engine. Write different processes. 2

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(b) Draw valve timing diagram of a four stroke SI engine. Why valves are open before TDC and closes after BDC ? 7

(c) Explain construction and working of Rotary Wankel engine with the help of neat sketch. 7

(d) Assuming fuel air cycle approximation, discuss the effect of fuel-air ratio on following parameters of otto cycle :

(i) Efficiency,

(ii) Maximum temperature

(iii) Maximum pressure

(iv) Power

Support your answers with suitable graphs. 7

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Q. 2. (a) What are the desirable properties of good IC engine fuel ? **2**

(b) What is ASTM distillation curve ? Write effect of volatility on : **7**

(i) Hot & Cold Starting

(ii) Carburettor icing

(c) Explain stages of combustion in CI engines. What is the importance of delay pereiode. **7**

(d) Orsat analysis of the products of combustion of hydrocarbon fuel of unknown composition is as follows : **7**

Carbon dioxide (CO_2) = 8%

Carbon monoxide (CO) = .5%

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Oxygen (O_2) = 6.3%

Nitrogen (N_2) = 85.2%

Determine :

(i) Air-Fuel ratio

(ii) Percent theoretical air required for combustion.

Q. 3. (a) Write drawback of simple carburettor. **2**

(b) Derive an expression for air-fuel ratio from simple carburettor taking compressibility in to account. **7**

(c) Define the term idling. Explain why a rich mixture is required for idling. Describe with suitable sketch idling system of modern carburettor. **7**

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(d) A simple carburettor under a certain condition delivers 5.45 kg/hr of petrol with an air-fuel ratio of 15. The fuel jet area is 2 mm² with coeff of discharge .75. If the tip of fuel jet is 0.635 cm above the level of petrol in the float chamber. Venturi throat coefficient of discharge is 0.8. Calculate :

(i) The venturi depression in cm of H_2O necessary to cause air and fuel to flow.

(ii) The venturi throat dia

(iii) The velocity of air across the venturi throat.

Density of air = 1.29 kg/m³

Specific gravity of petrol = .72. 7

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Q. 4. (a) Define the term lubrication.

(b) What do you understand by ignition

Describe battery ignition system with the he

of sketch.

(c) What are the three main types of fuel injector

Describe the automatic injector with the h

of neat sketch ?

(d) Write notes on :

(i) Dry sump lubrication

(ii) Comparison water and air cooling sy

Q. 5. (a) Name the methods of determining e

friction.

(b) Describe Morse Test. Write assumption

and limitation of Morse test.

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(c) Draw following performance curves of an SI engine and comment on nature of curve : 7

(i) Indicated power Vs. speed at full throttle.

(ii) Specific fuel consumption Vs. speed at full throttle. 7

(d) During the trial of single acting oil engine, cylinder diameter 200 mm stroke 280 mm working on two stroke cycle and firing energy cycle. the following observation were made.

Duration of trial = 1 hour

Total fuel used = 4.22 kg; Calorific value = 44670 kg/kg.; Proportion of hydrogen in fuel = 15%; Total number of revolution = 21000; Mean Effective pressure = 2.74 bar; Net brake load

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applied to brake of 1 m diameter = 600 N; Tota mass of cooling water circulated = 495 kg; Inle temp. of cooling water = 13°C; Outlet temp. o Cooling water = 38°C; Air used = 135 kg; Temp of air in test room = 20°C. Temp. of exhaus gases = 370 C; Cp_g = 1.005, $Cp_{(steam)}$ = 2.09 kJ/kg K

Calculate thermal efficiency and draw hea balance sheet 7

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